Analytical Review of Preprocessing Techniques for Offline Handwritten Character Recognition

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Abstract— **Preprocessing techniques are the first step in a character recognition system. This paper deals with the various preprocessing techniques involved in character recognition system with different kind of images ranges from simple handwritten form based documents and documents containing colored and complex background and varied intensities. Here, we are going to discuss all important preprocessing techniques like skew detection and correction, image enhancement techniques of contrast stretching, binarization, noise removal techniques, normalization and segmentation, morphological processing techniques.**

Keywords- Binarization;Character Recognition;Noise Removal; Normalization; Preprocessing Techniques; Segmentation.

I. INTRODUCTION

Character Recognition is nothing but machine simulation of human reading. It is also known as optical character recognition [1]. Character recognition system is a system which is used to recognize the characters in any document containing handwritten and machine printed text, graphics, videos etc. and converting them into digitized format into machine-readable or ASCII Codes . A character recognition system may also use neural network and artificial intelligence.

A. Phases for Character Recognition

The process of handwritten character recognition can be divided into phases as shown in the Fig. 1.

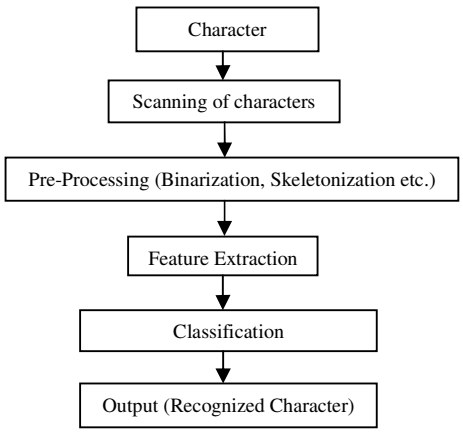


Figure 1. Phases of Character Recognition

1) Pre-processing: The process of extraction of text from the document is called pre-processing or document analysis. Pre-processing includes background noise reduction, filtering, original image restoration etc.

2) Feature Extraction: Feature extraction is the name given to a family of procedures for measuring the relevant shape information contained in a pattern so that the task of classifying the pattern is made easy by a formal procedure. Among the different design issues involved in building a recognizing system, perhaps the most significant one is the selection of set of features. Feature extraction for exploratory data projection enables high-dimensional data visualization for better data structure understanding and for cluster analysis. In feature extraction for classification, it is desirable to extract high discriminative reduced-dimensionality features, which reduce the classification computational requirements. Here, individual image glyph is considered and extracted for features. Each character glyph is defined by the following attributes: (1) Height of the character. (2) Width of the character. (3) Numbers of horizontal lines present short and long. (4) Numbers of vertical lines present short and long. (5) Numbers of circles present. (6) Numbers of horizontally oriented arcs. (7) Numbers of vertically oriented arcs. (8) Centroid of the image. (9) Position of the various features. (10) Pixels in the various regions.

3) Classification: Template matching or matrix matching, is one of the most common classification methods. Here individual image pixels are used as features. Classification is performed by comparing an input character with a set of templates (prototypes) from each character class. If the input characters matches with the (prototypes) templates character's identity is made equivalent to most similar template and similarity of measure increases; and if the input characters does not match with the template similarity of measure decreases [3]. All the above phases are applied in a pipelined fashion i.e. the success of each phase depend on the success of previous phase and all phases depend on the success and accuracy of each other phase as output of previous phase will be used as input to next phase.

Handwritten data is captured and stored in its digital format either by scanning the handwritten document by scanner or by writing with a stylus pen on an electronic

surface such as a digitizer/PDA or tablet with a liquid crystal displays. The two approaches may be distinguished as offline or online handwriting recognition [15]. A scanned paper may consist of either a machine printed/hand printed characters or handwritten characters. It may consist of images, graphics or any form kind data. Scanned paper forms included typewritten or computer-generated fonts, handwritten characters, check boxes and bubbles, bar codes, and signatures. A Data is to be extracted from these forms.

B. Application of Character Recognition

OCR has enabled scanned documents to become more than just image files, turning into fully searchable documents with text content that is recognized by computers. With the help of OCR, people no longer need to manually retype important documents when entering them into electronic databases. Instead, OCR extracts relevant information and enters it automatically. The result is accurate, efficient information processing in less time. The character recognition systems can be classified into 2 categories.

1) Task Specific Readers: The task specific readers handle only specific document types. Some of the most common task specific readers read bank check, letter mail, or credit card slips. This considerably reduces the image processing and text recognition time. Some application areas to which task specific readers have been applied are:

a) Form Readers: There are various form readers available.

- Legal Form Readers
- Healthcare Patient Forms Readers
- Signature Verification
- Stamp Recognition

b) Check Readers: OCR is used to process banking checks without human involvement. A check can be inserted into a machine, the writing on it is scanned instantly, and the correct amount of money is transferred. A check reader captures check images and recognizes courtesy amounts and account information on the checks. Some readers also recognize the legal amount on the check and use information in both fields to cross-check the recognition results.

c) Bill Processing Systems: In general a bill processing system is used to read payment slips, utility bills and inventory documents. The system focuses on certain region on a document where the expected information is located.

d) Airline Ticket Readers: In order to claim revenue from airline passenger ticket, an airline needs to have three records matched: reservation record, the travel agent record and the passenger ticket. However it is impossible to match all three records for every ticket sold. Several airlines are using a passenger revenue accounting system to account accurately for passenger revenue. The system reads the ticket number on a passenger ticket and matches it with the one in the airline reservation database. It scans tickets up to 260,000 tickets per day and achieves sorting rate of 17 tickets per second.

e) Handwritten Address Interpretation/ Address Readers: Handwritten address interpretation technology has also been developed for address recognition systems for the United Kingdom's Royal Mail and the Australia Post.

The address in a postal mail sorter locates the destination address block on a mail piece and read the ZIP code in the address book. If additional fields in the address block are read with high confidence the system may generate a 9 digit ZIP code is used to generate a bar code which is sprayed on the envelope.

2) General Purpose Page Readers: They are designed to handle a broad range of documents such as business letters, technical writing and newspapers. These systems capture an image of a document page and separate the page into text and non text region. Non-text regions such as graphics and line drawings are often saved separately from the text and associated recognition results. Text regions are segmented into lines, words and characters and the characters are passed to the recognizer. Recognition results are output in a format that can be post-processed by the application software. Most of these page readers can read machine written text only; a few can read hand-printed alphanumeric [3].

C. Limitations of Character Recognition System

OCR is unable to achieve a 100% recognition rate. Because of this, a system which permits fast and accurate recognition is a major requirement. The success of any OCR device to read accurately is the responsibility of the hardware manufacturer as well as depends on the quality of the items to be processed. The main purpose of OCR from many years is as follows:

- To increase the accuracy of recognizing.
- To eliminate the need for specially designed fonts (character).

In fact OCR is a time saver as it reduces the manual work, but it is not perfect.

- It hardly reaches more than 99.9% accuracy level.
- It faces problem with early printed books, newspaper etc.
- It faces problems with heavily bound material.

D. Handwritten Character Recognition

Handwriting character recognition is the software used to read, interpret and receive any texts in the handwritten form. The sources of such printed words may be from documents, images etc. The two approaches may be categorized as offline and online handwriting recognition as shown in Fig. 2.

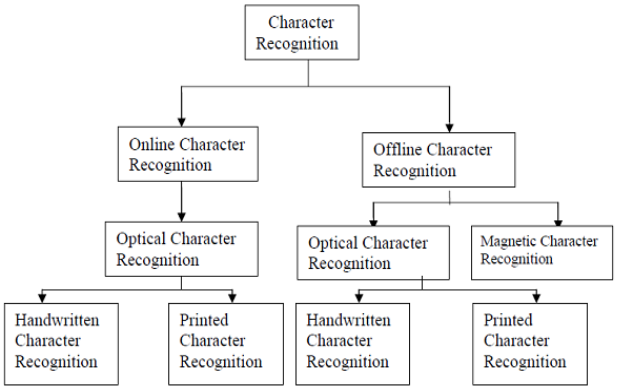


Figure 2. Character Recognition Classification

TABLE I. COMPARISON BETWEEN ONLINE AND OFFLINE HANDWRITTEN CHARACTERS

Sr No.	Comparisons	On-line characters	Off-line characters
1.	Availability of no. of pen-strokes	Yes	No
2.	Raw Data Requirement	# samples/second(e.g. 100)	# dots/inch(e.g. 300)
3.	Way of writing	Using digital pen on LCD surface	Paper document
4.	Recognition Rates	Higher	Lower
5.	Accuracy	Higher	Lower

Most of the hundreds of documents that a business deals come with handwritten texts. The handwriting character recognition is an assured way of getting these words correctly and converting them into digitized formats. Most of the traditional OCR would find it difficult because the handwritten texts vary from a person to a person. Unlike the printed fonts, they have definite sizes and shapes that are stored within the program library. The handwriting character recognition is the software that is advanced in terms of understanding and reading the different texts. A comparison between Offline and Online Handwritten Character Recognition is shown in Table I. Off-line Handwriting/Character recognition: Off-line handwriting recognition refers to the process of recognizing words that have been scanned from a surface (e.g. from a sheet of paper) and are stored digitally in grey scale format. After being stored, further processing on the image is done to allow good recognition. The offline character recognition can be further grouped into two types:

- Magnetic Character Recognition (MCR)
- Optical Character Recognition (OCR)

In MCR, the characters are printed with magnetic ink. The reading device can recognize the characters according to the unique magnetic field of each character. MCR is mostly used in banks for check authentication. OCR deals with the recognition of characters acquiring by optical means, typically a scanner or a camera. The characters are in the form of pixelized images, and can be either printed or, of any size, shape, or orientation. The OCR can be subdivided into character recognition and printed character recognition. Character Recognition is more difficult to implement than printed character recognition due to different human handwriting styles. In printed character recognition, the images to be processed are in the forms of standard fonts like Times New Roman, Arial, Courier, etc. The *drawbacks* of the off-line recognizers, compared to on-line recognizers are summarized as follows:

- Off-line conversion usually requires costly and imperfect pre-processing techniques prior to feature extraction and recognition stages.
- They do not carry temporal or dynamic information such as the number and order of pen-on and pen-off movements, the direction and speed of writing and in some cases, the pressure applied while writing a character.
- They are not real-time recognizers.

1) On-line Handwriting/Character recognition: In the On-line character recognition the process of recognizing handwriting is recorded with a digitizer as a time sequence of pen coordinates. In case of online character recognition, the handwriting is captured and stored in digital form via different means. Usually, a special pen is used in combination with an electronic surface. As the pen moves across the surface, the two- dimensional coordinates of successive points are represented as a function of time and are stored in order. Researches on online character recognition revealed that the on-line method of recognizing text has achieved better results than its off-line counterpart. On-line handwriting recognition involves the automatic conversion of text written on a special digitizer or PDA or tablet, where a sensor picks up the pen-tip movements as well as pen-up/pen-down switching the data obtained in this form is called one dimensional text. This may be attributed to the fact that more information may be captured in the on-line case such as the direction, speed and the order of strokes of the handwriting. The on-line handwriting recognition problem has a number of distinguishing features which must be used to get more accurate results for the online recognition problem.

- It is adaptive: The immediate feedback is given by the writer whose corrections can be used to further train the recognizer.
- It is a real time process: It captures the temporal or dynamic information of the writing. This information consists of the number of pen strokes (i.e. the writing from pen down to pen up), the order of pen-strokes. The direction of the writing for each pen stroke and the speed of the writing within each pen stroke.

II. PREPROCESSING TECHNIQUES

Preprocessing techniques are applied only on grey level and binary document images i.e. binary images containing text and/or graphics. In character recognition systems most of the applications use grey or binary images since processing color images is more difficult and time

consuming. Such images may also contain non-uniform background and/or watermarks making it difficult to extract the document text from the image without performing some kind of preprocessing, therefore; the desired result from preprocessing is a binary image containing text only.

A preprocessing stage of the grey scale source image is essential for the elimination of noisy areas, smoothing of background texture as well as contrast enhancement between background and text areas. OCR system can be made robust through applying correct Image enhancement techniques like, noise removal, image thresholding, skew detection/ correction, page segmentation, character segmentation, and character normalization and morphological techniques [2, 7]. Preprocessing techniques in detail are illustrated as shown in Fig. 3.

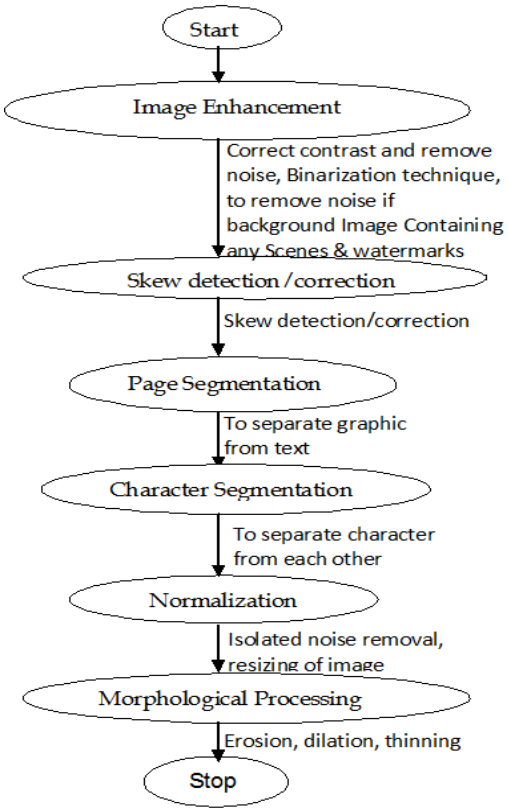


Figure 3. Flowchart Illustrating the Preprocessing techniques

A. *Image enhancement techniques*

The main objective of the image enhancement is to modify attributes of image to make it more suitable for a given task and improve the quality of image for the human perception by reducing noise, image blurring, increasing contrast and providing more details. The principle objective of image enhancement is to process an image so that result is more suitable than original image for specific application. Digital image enhancement technique provides a multitude of choices for improving the visual quality of image. Appropriate choice of such techniques is greatly influenced by imaging modality task at end. Image enhancement is basically improving the interoperatibility or perception of information in images for human viewers and providing better input for automated image processing techniques. Thresholding and mask processing techniques are also included in spatial domain of image enhancement techniques [2, 9, 11].

B. *Binarization (Thresholding)*

Document Image Binarization converts the image into bi-level form in such a way that foreground information is represented by black pixels and background by the white pixels Preprocessing techniques are applied only on grey or binary images, a grey image is one in which pixel density value is in between 0 and 255 and a binary image is one in which pixel density value is in the form of 0 and 1 where 0 stands for white i.e. background of image and 1 stands for black i.e. foreground of image. The process of conversion of a grey scaled image into binary image is known as binarization and the method used for binarization is known as Thresholding. In thresholding grey scale or color images are represented as binary images by picking a thresholding value. Fig. 4 shows the effects of applying thresholding operations on an image.

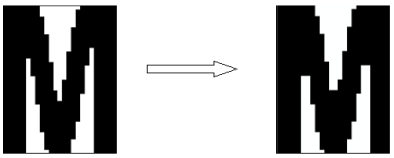


Figure 4. Applying thresholding operation on image

There are three kinds of Thresholding - Global thresholding, local thresholding and Hybrid thresholding. [4,13,15,16].

1) Global Thresholding: One threshold value is used for the entire document image which is often based on an estimation of the background level from the intensity histogram of the image. The main drawback of global thresholding is that it cannot adapt well to uneven illumination and random noises, and hence performs unsatisfactorily for low quality document images. e.g. Otsu's method, iterative method, valley point minimum method for simple images like handwriting, where the characters are written on a white background, using a global threshold would suffice to distinguish the background and foreground.

A thresholding application has to be performed on scanned gray scale images. Otsu's method is used in this work for the purpose of selecting the threshold and binarizing the gray scale images, so that resulting image has 0 as background pixels and 1 as foreground pixels.

a) Otsu's Method: This method can be explained using following steps:

Step 1: count the number o pixel according to color (256 color) and save it to matrix count.

Step 2: calculate probability matrix P of each color,

P_i = $count_i$ / sum of count, where i= 1, 2, … … 256.

Step 3: ind matrix omega, omegai = cumulative sum of P_i , where i= 1, 2,… 256.

Step 4: find matrix mu, mu_i = cumulative sum of P_i *I, where i= 1, 2, … …256 and mu_t = cumulative sum of 256 * 256

Step 5: calculate matrix sigma_b_squared where,

$$sigma_b_squared_i = \frac{(mu_t \times omega_i - mu_i)^2}{Omega_i-(1-Omega_i)}$$

Step 6: Find the location, idx, of the maximum value of sigma_b_squared. The maximum may extend over several bins, so average together the locations.

Step 7: If maximum is not a number, meaning that sigma_b_squared is all not a number, and then threshold is 0.

Step 8: If maximum is a finite number, threshold = (idx - 1) / (256 - 1);

In a grayscale image there are 256 combinations of black and white colors where 0 means pure black and 255 means pure white. This image is converted to binary image by checking whether or not each pixel value is greater than 255-level (level, found by Otsu's Method). If the pixel value is greater than or equal to 255-level then the value is set to 1 i.e. white otherwise 0 i.e. black.

2) Local or Adaptive Thresholding: Different values are used for each pixel according to the local area information of document image. In comparison to Global thresholding, local thresholding generally performs better for low quality document images. Local thresholding technique is used for degraded documents containing uneven illumination. This technique is commonly used in works that involve images that are of varying level of intensities, such as pictures from satellite cameras or medical scanned images. e.g. Otsu's adaptive methods, Novel technique, Niblack method, Fuzzy C means method, Interval Type -2 Fuzzy logic etc.

3) Hybrid Thresholding Technique: It attempts to combine the advantages of global and local thresholding, that is, better adaptability of various kinds of noise at different areas of the same image in low computational and time cost. Neural network technology is used by combining the results of best found and applying Kohonen Self organizing Map neural network [14] or/and combining results of global and local thresholding to obtain better results [13]. Image enhancement methods can broadly be divided into the following two categories:

a) Spatial Domain Techniques: In spatial domain techniques we directly deal with image pixel. The pixel values are manipulated to achieve desired enhancement. It is better than frequency domain as it does not require excessive computations. Using a small convolution mask such as 3×3, and convolving this mask over an image is much easier and faster than performing Fourier transform and multiplications.

b) Frequency (Compressed) Domain Technique: In frequency domain method image is first transferred into frequency domain, It means Fourier transform of image is computed first and then inverse Fourier transform performed first to get the resultant image. Fourier transform require substantial computations, and in some cases are not worth the effort. Multiplication in the frequency domain corresponds to convolution in time. E.g. DCT (Discrete cosine transform) for Morphological processing [2,9,11].

C. Noise Reduction Techniques

The major objective of noise removal is to remove any unwanted bit-patterns, which do not have any significance in the output. Noise reduction techniques are filtering, morphological operations and noise modeling. Filters can be designed for smoothing, sharpening, thresholding, removing slightly textured background and contrast adjustment process. Various morphological operations can be designed to connect broken strokes, decomposed the connected strokes, smooth the contours, clip the unwanted points, thin the characters and extract boundaries. Smoothing can be done by filters [6]. Types of filters available are:

1) Linear Filters (Averaging mask filter): Smoothing Low pass filter, Sharpening [2]. In grey level images low pass filter such as average or Gaussian blur filter proved to be best eliminating isolated pixel noise.

2) Non Linear Filters: Median filter is used as a non linear filter. In grey level images median filter proved to be best for eliminating isolated pixel noise. It is a low-pass filter. The median filter takes an area of an image (3x3, 5x5, 7x7, etc.), sorts out all the pixel values in that area, and replaces the center pixel with the median value. If the neighborhood under consideration contains an even number of pixels, the average of the two middle pixel values is used. The median filter is effective for removing impulse noise such as "salt and pepper noise" which is random occurrences of black and white pixels. The noise present in the image is removed by applying median filter. A median filter is more effective than convolution when the goal is to simultaneously reduce noise and preserve edges. Fig. 5 shows the original image and smooth image after applying filters.

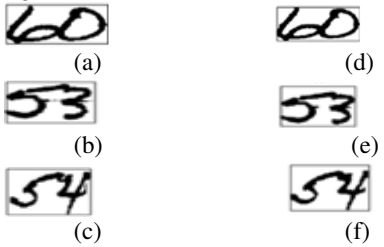


Figure 5. (a), (b), (c) original images and (d), (e), (f) are smoothed image

D. Skew detection and Correction

1) Skew: Deviation of the baseline of text from horizontal direction is called skew. It mainly concerns the orientation of text lines.

2) Skew Detection: When the pages are fed into the scanner they get skewed during scanning process. Skew detection affects the page segmentation/Classification phase. So skew detection is necessary for aligning text or document image so that text lines are aligned to co-ordinate axes. Techniques of skew detection can be

classified as under:

a) Detecting Connected Components- Finding the average angle connecting their centroid.

b) Projection Profile Analysis- A document page is projected at several angles and variances in the no. of black pixel per projected line are determined.

*c) Parallel Projection-*Each given ray projected through the image will hit either almost no black pixel (as it passes between text lines) and many black pixels (while passing through many characters in sequence).

*3) Skew Correction-*After the skew angle is detected, the page must be rotated in order to correct this skew. Rotation transformation algorithm is used to correct this skew. A co-ordinate rotation transformation can be used in equation given below.

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Given a point, its new co-ordinates after rotating the entire image around its origin by detecting skew angle of document image as shown in Fig. 6.

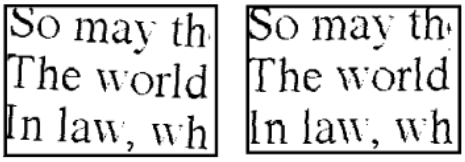


Figure 6. A Rotation transformation to correct skew of document image

E. Character Segmentation

It is considered as main step in cursive script such as Arabic, Urdu, etc. The techniques for character Segmentation is script specific and may not work well with other scripts. Even in printed handwritten documents segmentation is needed due to touching of characters. Connected pattern component are extracted and spatial interrelation between components are measured, and grouped into meaningful character patterns and 95% of connected characters are separated.

F. Image Normalization

Normalization is a linear process. If the intensity range of the image is 50 to 180 and the desired range is 0 to 255 the process entails subtracting 50 from each of pixel intensity, making the range 0 to 130. Then each pixel intensity is multiplied by 255/130, making the range 0 to 255. Auto-normalization in image processing software typically normalizes to the full dynamic range of the number system specified in the image file format. The normalization process will produce the regions, which have the same constant dimensions [30].Normalization methods aim to remove all types of variations during the writing and standardized data is obtained [26]. For example Size normalization is used to adjust the character size to a certain standard. Methods of character recognition may apply both horizontal and vertical size normalizations.

By applying (Neighborhood search based method to omit isolated noise blobs in character image while computing the image bounding box).Width and height of character are denoted by W_1 and H_1, the width and height of the normalized character are denoted by W_2 and H_2, and the size of standard plane is denoted by L. The size of standard plane is usually considered as a square and its size is typically 32× 32 or 64×64 among others. Aspect ratio R_1 and R_2 of original and normalized character are:

$$R_1 = \frac{\min(W_1, H_1)}{\max(W_1, H_1)},$$

$$R_2 = \frac{\min(W_2, H_2)}{\max(W_2, H_2)},$$

which are always considered in the range of [0,1].

G. Morphological processing

1) Thinning: It is a morphological operation that is used to remove the selected foreground pixels from binary images, somewhat like erosion or opening. Thinning is a data reduction process that erode an object until it is one pixel wide, producing a skeleton of the object making it easier to recognize characters.

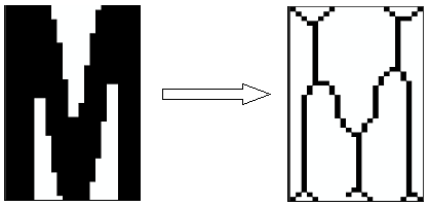


Figure 7. Applying Skeletonization on threshold image.

2) Erosion - Erosion makes an object smaller by removing or eroding away the pixels around its edges.

3) Dilation- Dilation makes an object larger by adding pixels around its edges.

*4) Closing-*It is similar to dilation. It tends to enlarge the boundaries of foreground region of image. It is less destructive of the original boundary shape. Closing joins one of broken objects and fills in unwanted holes in objects.

5) Opening- It is similar to erosion. It tends to remove some of the foreground pixels from the edges of regions of foreground. It is less destructive than erosion in general. Opening spaces objects that are too close together, detaches objects that are touching and should not be, and enlarges holes inside objects.

III. ANALYSIS

Different Techniques with their applications and ratings are described in table II-table VI.

TABLE II. DESCRIPTION OF POINT AND MASK PROCESSING TECHNIQUES.

Spatial Image filtering Operations/ Spatial domain Techniques	Application	Rating
Point Processing techniques	They are used to modify the pixel value in the original image to get a brighter and enhanced image and applications are contrast stretching, global thresholding, and histogram processing etc.	Good, Less costly
Mask Processing	The application of a mask to an input image produces an output image of same size as original image. Applications are noise removal, local thresholding, sharpening etc.	More Better and expensive

TABLE III. DESCRIPTION OF POINT PROCESSING TECHNIQUES.

Point Processing Techniques	Applications	Rating
Contrast, Stretching	Contrast stretching is used to increase or decrease the contrast to create a brighter or contrast adjusted and enhanced image. Applications are Piecewise linear contrast stretch, high pass and low pass nonlinear contrast stretch filter.	(Best) High pass and low pass filter
	Types of Contrast stretch : Linear stretch, Non-Linear Stretch	
	Types of Linear stretch: 1.Max-Min contrast stretch method 2.Percentage contrast stretch method 3.Piecewise contrast stretch Best is Piece wise contrast Stretch	
	Types of Non Linear contrast stretch: 1.Histogram equalization Types of histogram Equalization: • Local Histogram equalization • Global Histogram Equalization 2.Adaptive Histogram Equalization 3.Homographic filter Types of Homographic filter: • Low pass filter • High pass filter 4.Unsharp Mask Best is high pass and low pass filter	
Global image Thresholding	Uses only a single threshold value based on heuristics or statistics of global image attributes to classify image pixel into text (foreground) and non text pixel (background) the information of an image from its background applied to grey level or color document scanned images to reduce to a binary image. Otsu's class separability method/Otsu's method: Widely used technique to convert a grey level image into binary image then calculates optimum threshold separating those two classes so that their combined spread is minimal.	Otsu class separability method

TABLE IV. DESCRIPTION OF MASK PROCESSING TECHNIQUES

Mask Processing Techniques	Applications	Ratings
Smoothing Low pass filters	It can remove small pieces of noise salt and pepper noise in grey level image, remove unwanted details.	Better
Median filter	The median filter is effective for removing impulse noise such as "salt and pepper noise" which is random occurrences of black and white pixels.	Better
Local thresholding	Local thresholding is used in colored documents and where global thresholding fails we use local thresholding to separate object of interest from background. Foreground pixels are black and background are white or vice versa and separated for converting text to binary form. Niblack method, Bernsen's method, Local grey level thresholding, Sauvola & Pietikainen method etc.	

TABLE V. DESCRIPTION OF MORPHOLOGICAL OPERATIONS

Morphological operation	Applications	Rating
Erosion	Erosion makes an object smaller by removing or eroding away the pixels around its edges.	Good
Dilation	Dilation makes an object larger by adding pixels around its edges.	Good
Opening & Closing	**Opening:** Shape smoothing, deletion of small details at the object border, cutting of narrow items. **Closing:** Shape smoothing, filling up of narrow channels and small holes, connected components.	Better
Opening by reconstruction	In this method the background detection method is same as erosion, dilation method but the only thing the way to detect the background is modified.	Better
DCT	This technique is used to adjust color information in addition to brightness and contrast in color image.	Best

TABLE VI. DESCRIPTION OF NOISE REMOVAL, SKEW REMOVAL, NORMALIZATION, SEGMENTATION TECHNIQUES

Noise Removal	Forms are filled and mailed from all over. As a result they are received folded and are often dog-eared and smudged. Moreover, use of stapling pins, paper clips, salt and pepper noise etc introduce a lot of noise in the form image, Historical documents and degraded documents have uneven illumination, low contrast, random noise, isolated pixel noise about boundaries of documents. In forms there are checkboxes and to extract text boundaries are removed and functions and algorithms are used to remove this kind of noise. There are filters Weiner filter, averaging mask filters , median filter, shrink filter, swell filter, high pass and low pass filters, local registration and field box masking etc. to remove noise.	Correlation based algorithm (Good),Projection based algorithm, median filter, weiner filter (More Better).
Skew detection and correction	Skew and shift in document Like forms occurs during scanning process due to the possibility of rotation of the input image For e.g. In forms Skew detection techniques: (Global) form registration based robust and efficient algorithm, Hough transform analysis (Expensive), Nearest neighbor connection. Skew Correction technique- co-ordinate transformation rotation.	Better technique- Form registration based efficient and robust algorithm, co-ordinate transformation.
Page segmentation	It is used to separate text from halftone images, lines, and graphs. So that the result of document segmentation should be text only. Segmentation techniques: Top down, bottom up, hybrid techniques.	RLSA (Better)
Character Segmentation	Characters segmentation is needed in documents to remove touching of characters. Connected pattern component are extracted and spatial interrelation between components are measured, and grouped into meaningful character patterns and 95% of connected characters are separated. Techniques for character segmentation e.g. Printed Latin characters are easy to segment using horizontal and vertical histogram profile.	
Image Normalization	In it isolated noise in character image is removed, isolated noise blobs lead to inaccurate detection of bounding box, thus lead to inaccurate recognition. This may be due to the dust particle in ADF in automatic document feeder. Resizing of a character in order to reduce noise in image. Size is 32×32 or 64×64. Normalization is done .Linear, nonlinear and moment based normalization and neighborhood search based algorithm is used.	Neighborhood search based method to omit isolated noise blobs in character image while computing the image bounding box (Best).

IV. CONCLUSION

It is found that major activities in a typical preprocessing system are image enhancements, noise removal, contrast adjustment, binarization, normalization, segmentation. All these techniques are necessary in a preprocessing system; no technique alone cannot be said as complete activity for preprocessing e.g. only using noise removal we cannot completely preprocess an image. Therefore all image enhancement techniques have to be applied for accuracy in preprocessing system. Preprocessed image can be used in feature extraction phase and then in neural network phase. Even applying all these techniques we cannot obtain the 100% accuracy in a preprocessing system.

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